Software Requirements Specification
Obelisk

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Revision history

Version	Date	Changes
1.0	October 23, 2026	First release of the requirements.
1.1	October 23, 2026	Applied the design, data model, and security review fixes. Added decisions OD-13 to OD-21.
1.2	October 26, 2026	Added the Word export and .obk file libraries (docx, JSZip, yauzl, yazl) to the license reference and the software libraries table.
1.3	October 27, 2026	Recorded spikes S-01 to S-09 and owner decisions on decoding, elkjs, and diagram click links. Applied the approved menu wording and named the link handler owner.

1. Introduction

1.1 Purpose

This document defines what Obelisk must do and how well. Obelisk is a document editor for Windows, for one person on one PC. It writes pages, lays them out, and prints or exports them.

1.2 Intended audience

Filliciano Plong, the sole developer, uses this document to plan, build, and test Obelisk. Sections 6 and 7 hold the requirements; section 12 says how each is accepted.

1.3 Problem statement

Papers, letters, and problem sets need real pages, good math, and clean exports. Web editors need an account and a connection; full office suites are heavy and weak at math. Obelisk is a local editor that lays out real pages, makes equations easy, and exports to PDF, Word, and Markdown.

1.4 Product overview

Obelisk is an editor-only document processor. Each document is one .obk file in an ordinary Windows folder, and open documents sit in tabs. A page is built from blocks: text, headings, lists, tables, images, and equations. The equation block is the standout feature. Obelisk is built with Electron and released only through the Microsoft Store.

1.5 Definitions

Term	Meaning
Document	One .obk file: blocks, images, page setup, header, footer, and view.
Tab	One open document in the window. Settings and About also open as tabs.
Block	One piece of the page, such as a paragraph, table, image, or equation. Listed in the *Block types reference*.
Block handle	The grip left of a block, used to select, drag, and open the block menu.
Inline item	Something inside a line of text: a link, an inline equation, or a date and time item.
Pages view	Separate Letter pages with margins, headers, and footers. Always used for print and PDF.
Pageless view	One long page with no page edges. Headers and footers are hidden on screen.
Autosave	Writing a saved document to its own file shortly after each edit.
Recovery copy	A hidden copy of a never-saved tab, kept in app data in case of a crash.
Session	The list of open files kept in app data while Restore tabs on launch is on.
Manifest	The manifest.json entry of an .obk file: format name, version, and image list.
Slash menu	The block menu that opens when / is typed in an empty line.
Typed shortcut	Markdown-style characters that turn into formatting or a block as they are typed.
Equation block	A block of display math written in LaTeX and drawn by KaTeX.
Inline equation	Math inside a line of text, typed between $ signs.
Date and time item	An inline item holding one fixed date and time.
Column row	Two to four blocks side by side, each with a share of the text width.

Term	Meaning
Drawing layer	Strokes drawn over an image, kept with that image.
Signature block	A visual signature: an image of a signature with an optional line and labels. Not a verified digital signature.
App data	The per-user folder where Obelisk keeps settings, the session, signatures, and recovery copies. MSIX redirects it.
MSIX	The Windows app package format that the Microsoft Store installs and signs.
KaTeX	An open-source library that draws LaTeX math in HTML and MathML.
MathML, OMML	Markup for math: MathML on the web, OMML (Office Math) inside Word files.
GFM	GitHub Flavored Markdown, the Markdown dialect used for export and typed shortcuts.
LGPL, OFL	GNU Lesser General Public License; SIL Open Font License.
DPAPI	The Windows Data Protection API, used by Electron safeStorage on Windows.
MoSCoW	Priority scheme: Must, Should, Could. This document uses Must and Should.

1.6 Related documents

Lists of features, keys, and limits are in the named reference tables in the References section at the end.

- Obelisk architecture document (Plong_Obelisk_Architecture.docx), version 1.3.
- Electron documentation, safeStorage API: electronjs.org/docs/latest/api/safe-storage.
- Electron security checklist: electronjs.org/docs/latest/tutorial/security.
- KaTeX documentation, supported functions: katex.org/docs/supported.html.
- KaTeX documentation, options: katex.org/docs/options.html.
- Mermaid configuration schema: mermaid.js.org/config/schema-docs/config.html.
- mathml2omml on npm (version 0.5.0, LGPL-3.0-or-later): npmjs.com/package/mathml2omml.
- GitHub Flavored Markdown specification: github.github.com/gfm.
- Microsoft Store policies and app certification requirements: learn.microsoft.com/windows/apps/publish.
- WCAG 2.1: w3.org/TR/WCAG21.
- MLA Handbook, 9th edition (for the MLA template).

2. Scope

Obelisk is an editor only. Files live in Windows folders and open documents sit in tabs. Section 4 describes who uses it.

2.1 In scope

- Install Obelisk from the Microsoft Store and open .obk files from File Explorer.
- Create, open, edit, and save documents in any Windows folder.
- Work on several documents at once, one per tab.
- Lay out Letter pages with headers, footers, and page numbers, or write pageless.
- Format text, and build pages from blocks, tables, images, and equations.
- Move and copy blocks within a document, between tabs, and to other apps.
- Print, and export to PDF, Word, and Markdown.

- Should: date and time items, columns, drawing, signatures, diagrams, and crash recovery for never-saved tabs.

2.2 Out of scope

- A folder tree, starfield, or calendar view.
- Imports of any format, including Polaris .plr files.
- Templates other than MLA.
- A different first page for headers and footers.
- Settings for the default view, the undo depth, or Markdown shortcuts.
- Verified (cryptographic) digital signatures.
- Accounts, cloud sync, and collaboration.
- Releases outside the Microsoft Store, or for macOS or Linux.

Deferred items with reasons are in section 13.

3. Overall description

3.1 Product context

Obelisk is a standalone desktop app with no server. It uses Windows for file dialogs, printing, the clipboard, spell checking, and DPAPI. Files are ordinary files in the user's folders.

3.2 Main functions

- Tabs, menus, status bar, and Settings (6.1, 6.2).
- Files: open, check, save, autosave, and recovery (6.3).
- Page layout, formatting, and blocks (6.4 to 6.8).
- Tables, images, and equations (6.9 to 6.11).
- Find and Replace, print, and export (6.12, 6.13).
- Should features: date and time items, columns, drawing, signatures, diagrams (6.14 to 6.18).

3.3 Operating environment

- Windows 11 or Windows 10 22H2, x64 (OD-11).
- Installed from the Microsoft Store as an MSIX package for the current user.
- A printer driver, or Microsoft Print to PDF.
- Reference machine for targets: 4-core x64 laptop, 8 GB memory, SSD, 1920 by 1080, Windows 11.

4. User classes

Class	Role	Experience	Main tasks	Frequency
Student	Writes papers and problem sets to hand in	Uses a word processor; knows some LaTeX	MLA paper, equations, tables, PDF hand-in	Daily
Professional	Writes letters, briefs, and simple forms	Uses a word processor; no LaTeX	Signed letter, Word export for others to edit	Weekly

These are two example roles of one person, not two accounts. Either may use the keyboard only or Windows Narrator.

5. Constraints and assumptions

5.1 Constraints

Constraint	Source
Built with Electron, React, and TypeScript.	Developer decision
Released only through the Microsoft Store as an MSIX package.	Developer decision
Publishing needs a Partner Center account and a reserved app name (OD-01).	Microsoft Partner Center
Letter paper only; English interface.	Developer decision
Solo developer; scope is cut in the Should drop order if time runs short.	Project plan
Equations use KaTeX; Word equations use mathml2omml, which is LGPL.	Developer decision; library licenses
Bundled fonts: the LibreOffice Latin set plus Dancing Script (OD-08).	Developer decision

5.2 Assumptions

Assumption	How to validate
The user works on one Windows PC.	Confirm with the two user roles; no sync is asked for.
Word for Windows opens the exported .docx, including OMML equations.	Checked in Microsoft 365 Version 2610. Spike S-03 opened the six Equation reference samples as editable equations. Spike S-09 opened the Export reference rows, the same six samples as equations and as images, and a 50-page file.
GitHub renders the exported Markdown, including math and Mermaid.	Push a test export to a scratch repo.
Saving files to the user's folders is not redirected by MSIX.	Checked in spike S-01 on 27 October 2026, Windows 11 Home 10.0.26300, Electron 45.0.0, commit 8f4c177. Documents on that PC is OneDrive Known Folder Move, and a local Desktop folder was the second folder. Both writes landed on the real path. App data was redirected, and a document must not default to that folder. See architecture section 5.6.
The PC has a printer driver or Microsoft Print to PDF.	Default on Windows 10 and 11.

6. Functional requirements

Each line is one testable behavior with one MoSCoW priority. Must is required for release. Should is planned, and is cut in the order below if time runs short.

Verification is Test, Demo, Inspection, or Analysis. Lines flagged UNVERIFIED rest on a claim still to check.

Should drop order. Date and time items and columns are Shoulds outside this list.

1. Diagram block (6.18).
2. Drawing pad and tools, including drawing over images (6.16).
3. Signature block (6.17). If the drawing pad is already cut, signatures keep the typed and image modes.
4. Word equations (FR-57). The image fallback in FR-56 covers equations in Word.

6.1 Window and shell

Obelisk is one window. Open documents sit in tabs; there is no folder tree.

ID	Requirement	Priority	Verification
FR-01	The system shall show one window with a menu bar, a tab strip, the page, and a status bar.	Must	Demo: launch and point out each region.
FR-02	The system shall offer the menus and items in the *Menus reference*, each enabled only when it applies.	Must	Inspection: compare each menu with the reference.
FR-03	The system shall open each document in its own tab and mark a tab that has unsaved changes.	Must	Test: open three files, edit one, check its mark.
FR-04	The system shall show page, word, and character counts, zoom, and save state, including Autosave paused, in the status bar.	Must	Test: compare counts with a known sample; after a conflict Cancel, Autosave paused shows.

6.2 Settings

Settings hold only what the user may change. Undo depth, the default view, and Markdown shortcuts are fixed, not settings.

ID	Requirement	Priority	Verification
FR-05	The system shall open Settings in a tab, offering the options in the *Settings reference*.	Must	Inspection: compare the page with the reference.
FR-06	The system shall keep settings in app data and apply them again after a restart.	Must	Test: set Dark, restart, check the theme.

6.3 Files

Each document is one .obk file in an ordinary Windows folder. Saving is the user's choice until the first Save; after that, autosave keeps the file current.

ID	Requirement	Priority	Verification
FR-07	The system shall create a new, never-saved document in a new tab on File > New.	Must	Test: New opens an empty tab and writes no file.
FR-08	The system shall open a file chosen in the Windows Open dialog, which lists only Obelisk (.obk) files.	Must	Test: the dialog filter shows only .obk files.
FR-09	The system shall open nothing for a file failing the *File format reference* checks, leave it unchanged, and say why.	Must	Test: a renamed text file, a damaged file, a newer-version file, and each hostile sample stay byte-for-byte unchanged.
FR-10	The system shall write the open document to its own file on Save.	Must	Test: save, reopen, compare content.
FR-11	The system shall ask for a folder and name in the Windows Save dialog on Save as or first Save.	Must	Demo: Save as writes a new file; the old file is unchanged.
FR-12	The system shall get the user's confirmation before Save as or an export replaces an existing file.	Must	Test: Cancel at the prompt leaves the existing file unchanged.
FR-13	The system shall offer Overwrite, Save as, or Cancel before saving over a file another program changed.	Must	Test: change the file in another program, then save in Obelisk.
FR-14	The system shall autosave a changed document within 30 seconds of its last edit (target), once it has been saved.	Must	Test: a never-saved tab writes nothing; a saved one is on disk 30 s after an edit.
FR-15	The system shall skip an autosave and show the FR-13 choice when the file changed on disk.	Must	Test: change the file in another program, then edit in Obelisk.
FR-16	The system shall ask to save, discard, or cancel when a tab or the app closes with unsaved changes.	Must	Test: close a changed tab, and exit with one.
FR-17	The system shall store each document's view, Pages or Pageless, in its file and reopen it in that view.	Must	Test: save in Pageless, close, reopen.
FR-18	The system shall open a double-clicked .obk file in Obelisk, in a new tab if Obelisk is already running.	Should	Demo: double-click two files from File Explorer; one window, two tabs.
FR-19	The system shall keep a hidden recovery copy of each changed, never-saved tab in app data every 30 seconds.	Should	Inspection: the copy appears in app data within 30 s of an edit.
FR-20	The system shall offer each recovery copy at every launch, to restore or discard, until the user chooses.	Should	Test: end Obelisk from Task Manager, restart, close the prompt, restart; the copy is offered again.
FR-21	The system shall delete a tab's recovery copy when that tab is saved or discarded.	Should	Inspection: the copy is gone after Save and after Discard.

Checked in spike S-05 on 27 October 2026, Windows 11 Home 10.0.26300, Electron 45.0.0, commit 7ea3b12. For Save as (`.obk`) and Export (`.docx`), `dialog.showSaveDialog` is owned by the app window. The Windows replace prompt appeared and stayed until a click. That includes a name typed without the extension, when the filter supplied `.obk` or `.docx`.

Replace returned the existing path. Cancel returned canceled and left the file unchanged. The dialog did not write the file.

Obelisk relies on that Windows prompt for Save as and Export. It does not show a second prompt. Obelisk's own prompt stays the fallback for another platform, or if a later Windows build stops asking.

Checked in spike S-02 on 27 October 2026, Windows 11 Home 10.0.26300, Electron 45.0.0, commit 0f95f5f, package Obelisk.SpikeS02 0.0.1.0. A double-clicked file reached the process on a cold start through argv. A second double-click reached the running process through `app.requestSingleInstanceLock` and the `second-instance` event. One window stayed open.

Selecting three files and pressing Enter delivered three `second-instance` events, one path each. Open with, and the app execution alias, also put the path on argv.

The `second-instance` argv includes Chromium switches before the path. Main takes the file paths out of argv. It does not read a fixed index.

Explorer set the working directory to `C:\WINDOWS\System32`. A relative path is not resolved against that directory.

FR-18 keeps this design (ADR-009). The unused fallback was to drop FR-18, or to use an MSIX activation object. This run did not need that.
`Windows.ApplicationModel.AppInstance.GetActivatedEventArgs` is not on Electron's `app` object.

6.4 Page view and layout

Pages view shows separate Letter pages. Pageless view shows one long page with no page edges. Print and PDF always use the page layout.

ID	Requirement	Priority	Verification
FR-22	The system shall open every new document in Pages view, showing separate Letter pages.	Must	Test: File > New opens in Pages view.
FR-23	The system shall switch a document between Pages and Pageless from the View menu, marking it changed.	Must	Test: switch the view; the tab shows unsaved changes.
FR-24	The system shall set orientation and margins as listed in the *Page setup reference*, stored in the file.	Must	Test: each margin preset changes the text width; reopen keeps it.
FR-25	The system shall show an editable header and footer on every page, with the fields in the *Page setup reference*.	Must	Test: a three-page document numbers 1, 2, 3 on screen and in the PDF.
FR-26	The system shall start a new page at the cursor on Ctrl+Enter or Insert > Page break.	Must	Test: text after the break starts on the next page.
FR-27	The system shall create a new document from the MLA template, as set in the *MLA template reference*.	Must	Inspection: compare the new document with the reference.

6.5 Text formatting

Formatting applies to text in any block that holds text, including table cells.

ID	Requirement	Priority	Verification
FR-28	The system shall apply and clear the character formatting, fonts, and sizes in the *Formatting reference*.	Must	Test: apply each mark, then Clear formatting.
FR-29	The system shall set alignment, line spacing, and space	Must	Test: each value changes the

ID	Requirement	Priority	Verification
	before and after each paragraph, per the *Formatting reference*.		layout on screen and in the PDF.
FR-30	The system shall set left, first-line, and hanging indents per paragraph, per the *Formatting reference*.	Must	Test: a hanging indent of 0.5 in on a Works Cited entry.
FR-31	The system shall underline misspelled words and offer suggestions from the Windows spell checker.	Must	Demo: misspell a word, pick a suggestion from the text menu.

UNVERIFIED: FR-31 assumes Electron uses the Windows spell checker on Windows and returns suggestions to the context menu. Check: Electron spellchecker docs.

6.6 Blocks and typing

A document is a list of blocks. Each block has a handle on its left for selecting, dragging, and its menu.

ID	Requirement	Priority	Verification
FR-32	The system shall provide every block in the *Block types reference*, any number of each.	Must	Test: insert two of each block, save, reopen.
FR-33	The system shall show a block handle beside the block under the pointer or the cursor.	Must	Demo: move the pointer and the cursor over blocks.
FR-34	The system shall open the slash menu on / in an empty line, listing *Slash menu reference* items in order.	Must	Test: type /tab and choose Table.
FR-35	The system shall always apply the typed shortcuts in the *Keyboard reference*, following its rules.	Must	Test: type each shortcut; Ctrl+Z restores the typed characters.
FR-36	The system shall paste from other apps only what the *Paste reference* keeps.	Must	Test: paste a web page with a script, a style, and a remote image.
FR-37	The system shall undo and redo up to 100 steps per tab, kept through Save until the tab closes.	Must	Test: 101 edits, then 100 undos; Save keeps the history.

6.7 Block selection and clipboard

Blocks can be moved and copied as whole units, within a document, between tabs, and to other apps.

ID	Requirement	Priority	Verification
FR-38	The system shall select a block by its handle or Esc, and a range by Shift+clicking another handle.	Must	Test: select three blocks with Shift+click.
FR-39	The system shall cut, copy, and paste selected blocks with Ctrl+X, Ctrl+C, and Ctrl+V, including between tabs.	Must	Test: copy a table from one tab to another.
FR-40	The system shall duplicate the selected blocks directly below them on Ctrl+D.	Must	Test: duplicate two blocks; order is kept.
FR-41	The system shall put copied blocks on the Windows clipboard as rich text, with plain text as a fallback.	Must	Test: paste into Word (rich) and Notepad (plain).
FR-42	The system shall move selected blocks by dragging the handle or with Alt+Shift+Up and Alt+Shift+Down.	Must	Test: move a block up two places by keyboard.

UNVERIFIED: FR-41 assumes Word accepts the HTML clipboard format Electron writes. Check with current Word for Windows.

6.8 Context menus

Every block and inline item has a context menu, reachable by mouse and keyboard.

ID	Requirement	Priority	Verification
FR-43	The system shall open a context menu on right-click, Shift+F10, or the Menu key.	Must	Test: each way opens the same menu.
FR-44	The system shall show the menu items in the *Context menus reference*, following its composition rule.	Must	Inspection: compare each menu with the reference.

6.9 Tables

Table cells hold text. A cell that starts with = holds cell-reference arithmetic. There are no named functions.

ID	Requirement	Priority	Verification
FR-45	The system shall insert a table, add or delete rows and columns, and toggle a bold header row.	Must	Test: build a 3 by 4 table with a header row; insert a row above a referenced cell; the formula follows it.
FR-46	The system shall compute a cell starting with = as cell-reference arithmetic, per the *Table math reference*.	Must	Test: =B2+B3*2, =(B2-B3)/2, =B2^2, =[B2+B3]**2.
FR-47	The system shall show #DIV/0!, #REF!, or #CYCLE! in a cell whose formula cannot be computed.	Must	Test: =B2/0, a deleted referenced row (its reference becomes #REF!), and a self-reference.
FR-48	The system shall sort table rows by one column, ascending or descending, keeping the header row fixed.	Must	Test: sort numbers and text both ways; row 1 stays.

6.10 Images

Images are stored inside the document file, so the file is complete on its own.

ID	Requirement	Priority	Verification
FR-49	The system shall insert an image from a file or the clipboard, within the *Image reference* limits.	Must	Test: insert each type and an animated GIF; a 25 MB image and an 8001-pixel image are refused.
FR-50	The system shall resize an image by dragging a corner handle, keeping its aspect ratio.	Must	Test: drag a corner; the ratio is unchanged.
FR-51	The system shall rotate an image by quarter turns and flip it horizontally or vertically.	Must	Test: rotate, flip, save, reopen.
FR-52	The system shall store alt text for each image and write it into Word and Markdown exports.	Must	Inspection: alt text in the .docx and the .md.

Checked in spike S-06 (Electron 45.0.0, Windows image 20261021.278.1, Ubuntu image 20261023.334.3, CI run 40461995542). Frame counts and frame sizes for GIF and WebP, including animated WebP, are read from the GIF89a blocks and the WebP RIFF chunks before any pixel decode. The first frame used for Word, PDF, and print is a PNG made in the sandboxed renderer from `ImageDecoder` frame 0. See architecture section 6.6.

6.11 Equations

The equation block is the standout feature: LaTeX math that renders as typed and exports cleanly.

ID	Requirement	Priority	Verification
FR-53	The system shall render LaTeX math with KaTeX in equation blocks and inline equations, per the *Equation reference*.	Must	Test: the sample equations in the reference render.
FR-54	The system shall show the rendered equation live below its LaTeX source while the equation is edited.	Must	Demo: type a fraction and watch it render.
FR-55	The system shall show KaTeX's error message in place of an equation that does not parse, still allowing Save.	Must	Test: \frac{1} shows an error and the file saves.
FR-56	The system shall export each equation to Word as an image when equation conversion is not built or fails.	Must	Test: with conversion off, equations appear as images in Word.
FR-57	The system shall export equations to Word as editable equations, converting KaTeX MathML to OMML with mathml2omml.	Should	Test: open the .docx in Word and edit an equation.

Checked in spike S-03 (mathml2omml 0.5.0, KaTeX 0.19.1, docx 9.14.0, Microsoft 365 Version 2610 on Windows, 27 October 2026). The six Equation reference samples open in Word as editable equations, display and inline. FR-57 keeps mathml2omml 0.5.0.

The exporter still cleans the OMML before it is written.

The sum's n-ary body is empty, and Word shows an empty placeholder with the summand beside it. The following sibling moves into that body.

Matrix parentheses are plain math runs and do not stretch. Stretchy fences around a matrix, and stretchy `mo` fences in general, become a delimiter with `begChr` and `endChr`.

Display equations are wrapped in `m:oMathPara`. In one sample the display matrix sat at the right edge and the closing parenthesis was cut off.

The dropped thin space in the integral, and the combining circumflex on theta, looked right. They stay as they are.

A formula whose MathML contains < uses `mml2omml` with `disableDecode` set, so the XML stays well-formed.

`ImportedXmlComponent.fromXmlString()` in docx 9.14.0 returns a document node. Inserting that node writes an element named `undefined`, and Word refuses the file. The `m:oMath` child is what goes in the paragraph. Spike S-09 saw the same refusal.

Alignment is still unchecked. The Equation reference has no alignment sample.

FR-56 stays the image fallback when a conversion fails.

6.12 Find and Replace

Find and Replace works on the active tab only.

ID	Requirement	Priority	Verification
FR-58	The system shall open Find or Replace in one bar, with Match case and Whole word options.	Must	Test: each option narrows the matches.
FR-59	The system shall highlight every match, step through matches, and show the match count.	Must	Test: a word used five times shows 5.
FR-60	The system shall replace the current match or all matches, with Replace all as one undo step.	Must	Test: Replace all, then one Ctrl+Z restores every match.

6.13 Print and export

Print and every export use the document as it is on screen, including unsaved changes. HTML and plain-text exports are future features.

ID	Requirement	Priority	Verification
FR-61	The system shall print through the Windows print dialog, laid out as Pages view, including unsaved changes.	Must	Demo: print to Microsoft Print to PDF.
FR-62	The system shall export a PDF laid out as Pages view, whichever view is shown.	Must	Test: export from Pageless; the PDF has pages, headers, and footers.
FR-63	The system shall export a Word (.docx) file, mapping each block as listed in the *Export reference*.	Must	Test: open the export in Word; check each row of the reference.
FR-64	The system shall export GitHub Flavored Markdown (.md), mapping each block as listed in the *Export reference*.	Must	Test: view the export on GitHub; check each row of the reference.
FR-65	The system shall write a Markdown export's images to a folder beside the .md file, linked by relative paths.	Must	Test: move the .md and its folder together; images still show. An existing folder is named in the replace prompt.
FR-66	The system shall leave the document file unchanged when printing or exporting.	Must	Test: compare the .obk bytes before and after.

6.14 Date and time

A date and time item is an inline item that holds one date and time. It never updates by itself.

ID	Requirement	Priority	Verification
FR-67	The system shall insert an inline date and time item, set with a calendar and clock picker and editable later.	Should	Test: insert an item, change its date twice.
FR-68	The system shall show each date and time item in a	Should	Test: each format shows the

ID	Requirement	Priority	Verification
	format from the *Date formats reference*, Windows by default.		example in the reference.
FR-69	The system shall keep a date and time item's value fixed; it never changes to the current date by itself.	Should	Test: reopen the file a day later; the item is unchanged.

6.15 Columns

Columns place blocks side by side in a column row. They exist only on the page; Markdown stacks them.

ID	Requirement	Priority	Verification
FR-70	The system shall create columns when a block is dropped on another block's left or right edge, or via /column.	Should	Test: drag an image onto a text block's right edge.
FR-71	The system shall split a column row evenly into two to four columns, never nesting columns.	Should	Test: a fifth column and a column inside a column are refused.
FR-72	The system shall resize column shares by dragging the gap, within the *Columns reference* limits.	Should	Test: drag to the minimum share; it stops there.
FR-73	The system shall turn a column row back into a plain block when only one column remains.	Should	Test: drag one of two blocks out of the row.
FR-74	The system shall lay out column rows across pages by the paging rules in the *Columns reference*.	Should	Test: a row near a page end moves whole; a very tall row splits.

6.16 Drawing

Drawing covers the drawing pad block and drawing over an image.

ID	Requirement	Priority	Verification
FR-75	The system shall provide a drawing pad block with the tools in the *Drawing tools reference*.	Should	Test: use each tool, save, reopen.
FR-76	The system shall draw over an image on a layer that follows it through resize, rotate, and flip.	Should	Test: draw an arrow, then resize, rotate, and flip the image.
FR-77	The system shall flatten an image's drawing layer into the image when exporting to PDF or Word.	Should	Inspection: the export holds one image with the drawing.

6.17 Signature block

A signature block is a visual signature: an image of a signature on a page. It is not a verified digital signature.

ID	Requirement	Priority	Verification
FR-78	The system shall insert a signature block via /signature, created by drawing, typing in a script font, or an image.	Should	Test: create one signature of each kind.
FR-79	The system shall save signatures for reuse in app data, encrypted as stated in NFR-08.	Should	Test: reuse a saved signature in a new document.

ID	Requirement	Priority	Verification
FR-80	The system shall offer the line, width, label, alignment, and blank-mode settings in the *Signature options reference*.	Should	Inspection: compare the settings with the reference.
FR-81	The system shall resize a signature and export it as an image in PDF, Word, and Markdown.	Should	Test: resize, then export to each format.

6.18 Diagram block

A diagram block draws a diagram from short text, for example a flowchart.

ID	Requirement	Priority	Verification
FR-82	The system shall render a diagram block from Mermaid text offline, showing the source while editing.	Should	Test: a flowchart renders with the network off.
FR-83	The system shall show an error message in place of a diagram whose source does not parse.	Should	Test: a broken source shows the message.

Spike S-04 checked Mermaid 12.2.0. The license is MIT. The renderer bundle was 12,325,210 bytes, plus 1,076,572 bytes of KaTeX fonts. A sandboxed Electron 45 renderer made no network request.

A readable flowchart uses a post-process: style attributes move into the diagram stylesheet, only drawing properties are kept, every selector is prefixed with the diagram id, and one style element is allowed with a per-render nonce. `style-src-elem` is `app:` plus that nonce. `style-src-attr` is `'none'`. There is no `'unsafe-inline'`.

APPROVED 27 October 2026. Diagrams use Mermaid's dagre layout. That layout is MIT. elkjs is not shipped, and there is no ELK layout option. Spike S-04 left the layout unset, and Mermaid 12.2.0 then used elk. That choice is not kept.

APPROVED 27 October 2026. Before a diagram is rendered, `click` lines and `href` values are removed from its source. A link cannot be left on the picture.

7. Non-functional requirements

Each line has a condition, a measure, one priority, and a verification method. Numbers are targets until measured.

7.1 Performance

Numbers are targets on the reference machine in section 5, not yet measured.

ID	Category	Requirement	Priority	Verification
NFR-01	Performance	The system shall show an editable new tab within 3 seconds of launch (target).	Must	Test: timed on the reference machine.
NFR-02	Performance	The system shall show a typed character within 50 ms in a 50-page document (target).	Must	Test: input-latency trace.
NFR-03	Performance	The system shall open or save a 50-page, 10 MB document within 1 second (target).	Must	Test: timed on a local SSD.
NFR-04	Performance	The system shall export a 50-page document to PDF, Word, or Markdown within 10 seconds (target).	Must	Test: timed for each format.

If an export fails or stops at its 60-second limit, Obelisk says why and offers Retry. It leaves no partial file, and a file it was replacing keeps its old bytes.

7.2 Security and privacy

ID	Category	Requirement	Priority	Verification
NFR-05	Security	The system shall keep all documents on this PC, open no listening port, and send no usage data.	Must	Analysis: packet capture during editing, spelling, formulas, a diagram, save, and every export shows no traffic.
NFR-06	Security	The system shall never run scripts from document content or load remote content.	Must	Test: a file and a paste holding a script and a remote image send no request.
NFR-07	Security	The system shall build the page only from the schema in the *File format reference*, never from stored HTML.	Must	Test: the hostile sample set (architecture section 6.10) under Node opens nothing or opens safely.
NFR-08	Security	The system shall encrypt each saved signature with Electron safeStorage before writing it to app data.	Should	Inspection: the stored signature file holds no readable image or text.
NFR-09	Security	The system shall save or show no signature, and say so, while safeStorage reports encryption unavailable.	Should	Test: with encryption reported unavailable or temporarily unavailable, nothing is written or deleted.

On Windows, safeStorage protects saved signatures with DPAPI. This protects them from other Windows users on the PC, not from other apps run by the same user (Electron safeStorage docs).

Checked in spike S-07 on 27 October 2026, Windows 11 Home 10.0.26300, Electron 45.0.0, commit a9f81193dc9a9e668a11510128986084c26bb1cb, package Obelisk.SpikeS07 0.0.1.0. Obelisk uses the async safeStorage API. The sync methods are deprecated in Electron 45 and due for removal in Electron 46.

DPAPI inside the MSIX package encrypted and decrypted the signature, including after a restart. The ciphertext held no readable name or image. `shouldReEncrypt` was false.

While encryption is unavailable, nothing is saved or deleted (NFR-09). Check `isEncryptionAvailable` before saving a signature. `isAsyncEncryptionAvailable` stayed true even when the key was bad, so it is not the unavailable check.

A stored key that does not start with `DPAPI` makes `isEncryptionAvailable` false, and encrypt fails too. A bad key otherwise shows up only as `Error while decrypting the ciphertext provided to safeStorage.decryptStringAsync.` The "temporarily unavailable" string did not appear. Treat any decrypt failure as signature unavailable: save nothing and delete nothing.

7.3 Reliability

ID	Category	Requirement	Priority	Verification
NFR-10	Reliability	The system shall leave a save target holding its previous bytes or the complete new file, never part of one.	Must	Test: kill the app during a large save, 20 runs.
NFR-11	Reliability	The system shall treat a write not finished within 5 seconds as not saved and keep the tab marked changed.	Must	Test: save to a stalling test folder.
NFR-12	Reliability	The system shall report a save as done only after the complete file has replaced its target.	Must	Test: the tab mark clears only after the replace.
NFR-13	Reliability	The system shall lose at most 30 seconds of edits to a saved document when the app or Windows stops.	Must	Test: edit for 40 s, end the process, reopen.
NFR-14	Reliability	The system shall remove its own leftover temporary save files the next time it saves to that folder.	Must	Test: leave a temp file, save again, check the folder.

7.4 Usability and accessibility

ID	Category	Requirement	Priority	Verification
NFR-15	Usability	The system shall make every action reachable by keyboard, with the keys in the *Keyboard reference*.	Must	Test: run UC-02 and UC-05 by keyboard only.
NFR-16	Usability	The system shall give every control, menu item, tab, and block a name and role that Narrator	Must	Test: Narrator walk-through of the window.

ID	Category	Requirement	Priority	Verification
		reads.		
NFR-17	Usability	The system shall meet WCAG 2.1 AA contrast in Light and Dark, and follow Windows contrast themes.	Must	Analysis: contrast checker on both themes; Demo: a contrast theme.
NFR-18	Usability	The system shall show no clipped or overlapping text at 100% to 200% scaling in a 1280 by 800 window.	Must	Inspection: screenshots at 100%, 150%, and 200%.
NFR-19	Usability	The system shall never show save state, selection, or errors by color alone.	Must	Inspection: each state also has text or a shape.
NFR-20	Usability	The system shall stop interface animations when Windows animation effects are off.	Should	Demo: turn animations off in Windows Settings.

7.5 Maintainability

ID	Category	Requirement	Priority	Verification
NFR-21	Maintain.	The system shall keep the file format, save rules, table math, and exporters in modules testable under Node.	Must	Inspection: those tests run with no Electron window.
NFR-22	Maintain.	The system shall touch the disk only from the main process; renderers reach it through a typed IPC bridge.	Must	Inspection: code review of renderer imports.
NFR-23	Maintain.	The system shall cover the file format, save, and table math modules with at least 80% line coverage (target).	Should	Analysis: coverage report.

7.6 Portability and packaging

ID	Category	Requirement	Priority	Verification
NFR-24	Portability	The system shall install from the Microsoft Store for the current user, without administrator rights.	Must	Test: install as a standard user.
NFR-25	Portability	The system shall run on Windows 11 and Windows 10 22H2, x64.	Must	Test: smoke test on both versions.
NFR-26	Portability	The system shall write nothing to its install folder, keeping settings, session, signatures, and recovery copies in app data.	Must	Inspection: file-access log during a session.
NFR-27	Portability	The system shall leave every document in place when uninstalled, and update only through the Microsoft Store.	Must	Test: uninstall; documents remain.

UNVERIFIED: The current Microsoft Store minimum Windows version for new MSIX apps (OD-11). Check: Microsoft Store policies.

7.7 Compliance

ID	Category	Requirement	Priority	Verification
NFR-28	Compliance	The system shall show an About page with the app version and every notice in the *Open-source license reference*.	Must	Inspection: each listed component has its notice and full license text.
NFR-29	Compliance	The system shall ship mathml2omml under its LGPL terms, as a separate replaceable module with its source offered.	Must	Inspection: the package holds the module unbundled; About links to its source.
NFR-30	Compliance	The system shall bundle only fonts whose licenses appear in the *Open-source license reference*.	Must	Inspection: each font file matches a row in the reference.
NFR-31	Compliance	The system shall pass Microsoft Store certification with a listing, a privacy policy URL, and an age rating.	Must	Demo: the certification report.

UNVERIFIED: What LGPL-3.0 requires for mathml2omml packed in an Electron app, and each font license. See the flags in the *Open-source license reference* and the *Fonts reference*.

8. Use cases

One use case per key user goal. Should use cases cover Should features and are marked in their names.

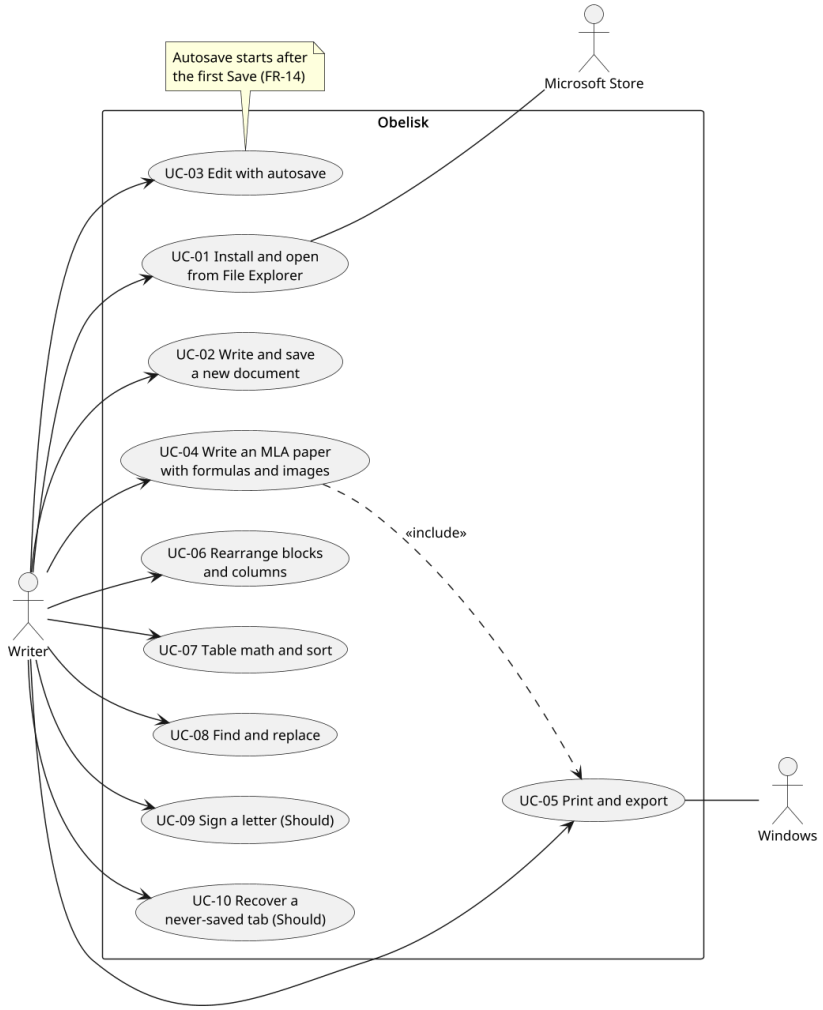


Figure 1. *Obelisk use cases*

Purpose: Shows the user goals covered by the use cases below.

Assumption: One user on one PC, in either role from section 4.

Limitation: Shows goals only; steps and errors are in the tables.

UC-01: Install Obelisk and open a file from File Explorer (Should)

Field	Content
Primary actor	Student or Professional
Goal	Get Obelisk and open a document from a folder.
Preconditions	Windows 11 or Windows 10 22H2; a Microsoft account in the Store.
Main flow	1. User installs Obelisk from the Microsoft Store.

Field	Content
	2. Store installs it for the current user, with no administrator prompt. 3. User double-clicks an .obk file in File Explorer. 4. System opens Obelisk with the file in a tab, in its stored view. 5. User double-clicks a second .obk file. 6. System opens it in a new tab of the same window. 7. User opens Settings and picks the Dark theme.
Expected result	Two documents open in one window, in the Dark theme.
Alternatives / errors	3a. The file fails the checks: no tab opens and a message says why. 3b. A newer Obelisk made the file: nothing opens; the message says to update from the Microsoft Store. 5a. The file is already open: system switches to its tab.
Related requirements	FR-05, FR-06, FR-09, FR-17, FR-18, NFR-24, NFR-25

UC-02: Write and save a new document

Field	Content
Primary actor	Student or Professional
Goal	Write a new document and keep it.
Preconditions	Obelisk is open.
Main flow	1. User chooses File > New. 2. System opens an empty, never-saved tab in Pages view. 3. User types headings and paragraphs, using typed shortcuts and the slash menu. 4. User presses Ctrl+S. 5. System shows the Windows Save dialog. 6. User picks a folder and a name. 7. System writes the file and clears the tab's unsaved mark.
Expected result	A complete .obk file in the chosen folder; autosave is now on for it.
Alternatives / errors	6a. A file with that name exists: system asks before replacing it. 7a. The write fails or passes 5 seconds: the tab stays never-saved, a message says why, and Retry is offered when it may help.
Related requirements	FR-01, FR-02, FR-03, FR-04, FR-07, FR-10, FR-11, FR-12, FR-22, FR-32, FR-33, FR-34, FR-35, FR-37, NFR-11, NFR-12

UC-03: Open a document and edit it with autosave

Field	Content
Primary actor	Student or Professional
Goal	Continue work on a saved document without thinking about saving.
Preconditions	A saved .obk file exists.
Main flow	1. User presses Ctrl+O and picks the file.

Field	Content
	2. System checks the file and opens it in a tab. 3. User edits text and formatting. 4. System autosaves within 30 seconds of the last edit. 5. User switches the document to Pageless view. 6. System marks the document changed and autosaves the new view. 7. User closes the tab.
Expected result	The file holds every edit and reopens in Pageless view.
Alternatives / errors	2a. The file fails the checks: nothing opens; the file is unchanged. 4a. Another program changed the file: system skips the autosave and offers Overwrite, Save as, or Cancel. After Cancel, the status bar shows Autosave paused until the next save. 7a. Unsaved changes remain: system asks to save, discard, or cancel.
Related requirements	FR-04, FR-08, FR-09, FR-13, FR-14, FR-15, FR-16, FR-17, FR-23, FR-28, FR-29, NFR-10, NFR-13

UC-04: Write an MLA paper with equations and images

Field	Content
Primary actor	Student
Goal	Hand in a correctly formatted paper with math and figures.
Preconditions	Obelisk is open.
Main flow	1. User chooses File > New from template > MLA. 2. System creates a document with MLA margins, font, spacing, header, and heading lines. 3. User writes the paper, adding equation blocks and inline equations. 4. System renders each equation live as it is typed. 5. User inserts an image, resizes it, adds alt text, and draws an arrow on it. 6. User adds a Works Cited list with hanging indents. 7. User exports a PDF.
Expected result	A PDF with MLA layout, page-numbered headers, rendered equations, and the annotated image.
Alternatives / errors	3a. An equation does not parse: system shows the KaTeX error in place; Save still works. 5a. The image is over the size limit: system refuses it and says why.
Related requirements	FR-24, FR-25, FR-26, FR-27, FR-30, FR-31, FR-49, FR-50, FR-52, FR-53, FR-54, FR-55, FR-62, FR-76

UC-05: Print and export a document

Field	Content
Primary actor	Student or Professional

Field	Content
Goal	Share the document as paper, PDF, Word, or Markdown.
Preconditions	A document is open, saved or not.
Main flow	1. User presses Ctrl+P. 2. System shows the Windows print dialog with the page layout. 3. User exports to Word. 4. System writes the .docx with equations as editable equations. 5. User exports to Markdown. 6. System writes the .md and an images folder beside it.
Expected result	Printed pages, a .docx, and a .md with images; the .obk file is unchanged.
Alternatives / errors	4a. Equation conversion fails or is not built: equations go into Word as images. 4b. The target file exists: system asks before replacing it.
Related requirements	FR-12, FR-56, FR-57, FR-61, FR-62, FR-63, FR-64, FR-65, FR-66, NFR-04

UC-06: Rearrange blocks and columns

Field	Content
Primary actor	Student or Professional
Goal	Reorder and reuse parts of a document.
Preconditions	A document with several blocks is open.
Main flow	1. User selects a block by its handle and Shift+clicks a second handle. 2. System selects the range of blocks. 3. User presses Alt+Shift+Up twice. 4. System moves the blocks up two places. 5. User copies the blocks and pastes them into another tab. 6. User drags an image onto the right edge of a paragraph. 7. System places them side by side in a column row.
Expected result	Blocks reordered, copied to the second tab, and an image beside text.
Alternatives / errors	6a. The row already has four columns: system refuses the drop. Any step: Ctrl+Z undoes the last move as one step.
Related requirements	FR-36, FR-37, FR-38, FR-39, FR-40, FR-41, FR-42, FR-43, FR-44, FR-70, FR-71, FR-72

UC-07: Build a table with cell math and sort it

Field	Content
Primary actor	Student or Professional
Goal	Total a small table of numbers and order it.
Preconditions	A document is open.
Main flow	1. User types /table and chooses Table. 2. System inserts a 3 by 3 table with a header row. 3. User enters numbers, and =B2+B3+B4 in a total cell.

Field	Content
	4. System shows the computed value, and the formula while the cell is edited. 5. User sorts by column B, descending. 6. System sorts the rows under the header row in one undo step.
Expected result	A sorted table with a correct total.
Alternatives / errors	3a. The formula divides by zero, points outside the table, or refers to itself: the cell shows #DIV/0!, #REF!, or #CYCLE!. 3b. User types a formula that does not parse, such as =B2+: the cell keeps it as text with an error underline.
Related requirements	FR-34, FR-45, FR-46, FR-47, FR-48

UC-08: Find and replace text

Field	Content
Primary actor	Student or Professional
Goal	Change a word throughout a document.
Preconditions	A document is open.
Main flow	1. User presses Ctrl+H. 2. System opens the Replace bar. 3. User types the word and its replacement, and turns on Whole word. 4. System highlights each match and shows the count. 5. User chooses Replace all. 6. System replaces every match as one undo step.
Expected result	Every whole-word match replaced; one Ctrl+Z restores them all.
Alternatives / errors	4a. No matches: the count shows 0 and Replace is disabled.
Related requirements	FR-58, FR-59, FR-60

UC-09: Sign a letter (Should)

Field	Content
Primary actor	Professional
Goal	Add a visual signature to a letter and send it as a PDF.
Preconditions	A letter is open.
Main flow	1. User types /signature. 2. System inserts a signature block and opens the signature pad. 3. User draws a signature and chooses Save for reuse. 4. System encrypts and stores the signature in app data. 5. User sets a signature line with a name label. 6. User inserts a date and time item beside it, then exports a PDF.
Expected result	A PDF with the signature image above a labeled line, and the date. The letter holds the signature as an ordinary image.

Field	Content
Alternatives / errors	3a. User types a name instead: system shows it in Dancing Script. 4a. Encryption is unavailable: system saves no signature and says so; the block still works. 4b. Saved signatures are temporarily unavailable: none are shown or deleted; the user can draw or type one.
Related requirements	FR-67, FR-75, FR-78, FR-79, FR-80, FR-81, NFR-08, NFR-09

UC-10: Recover a never-saved tab after a crash (Should)

Field	Content
Primary actor	Student or Professional
Goal	Get back work that was never saved when Obelisk stopped.
Preconditions	A never-saved tab has changes; Obelisk or Windows stopped.
Main flow	1. User starts Obelisk again. 2. System lists each recovery copy, with its first line and time. 3. User chooses Restore. 4. System opens the copy in a new never-saved tab. 5. User saves it. 6. System deletes the recovery copy.
Expected result	The work is back in a saved file; no recovery copy remains.
Alternatives / errors	2a. User closes the prompt: the copies stay and are offered at the next launch. 3a. User chooses Discard: system deletes the copy. 3b. A copy fails the file checks: only Discard is offered.
Related requirements	FR-09, FR-19, FR-20, FR-21

9. Data requirements

Entity	Fields	Stored in	Retention	Privacy
Document (.obk file)	Manifest (format name, version, image list), page setup, header, footer, view, blocks, images	User's folder	Kept until the user deletes it	Personal
Block	ID (unique; new on paste and duplicate), type, content, formatting; per type in the *Block types reference*	Inside the document	With the document	Personal
Image	Bytes named by content hash, type, pixel size, display width, alt text, rotation, flip, drawing layer	Inside the document	With the document; dropped on save once no block uses it	Personal
Settings	Version, theme, spelling, and the rest of the *Settings reference*	App data	Until uninstall	Settings

Entity	Fields	Stored in	Retention	Privacy
Session	Version, open file paths in tab order, active tab; never-saved tabs are not listed	App data	Replaced each session; deleted when Restore tabs is turned off	Personal (file paths)
Saved signature	Name, PNG image, created date; one encrypted file each	App data	Until deleted or uninstall	Sensitive
Recovery copy	A format 1 .obk file, plus created time, updated time, and first line	App data	Until saved or discarded; if ignored, kept and offered at every launch	Personal

- Not stored in the document: zoom, selection, find text, scroll position, and undo history. Open tabs are kept only in the session.
- Units: page sizes, margins, indents, and displayed image and signature widths in inches; font sizes and spacing in points; an image's own size in pixels. Image height is not stored; it follows the width and the aspect ratio.
- Table cells: an empty cell is empty, not zero; table math counts it as 0 only inside a formula.
- Dates in date and time items are stored as local date and time to the minute (YYYY-MM-DDTHH:MM), with no time zone, because the value never changes. The format is stored separately.
- A signature placed in a document is stored in that document like any image, not encrypted. Deleting a saved signature leaves documents that use it unchanged.
- Recovery copies are not encrypted, even when the tab holds a signature. They are the user's own content in the per-user app data folder.
- App data files are checked when read, like any file. A bad preference value falls back to its default; a damaged recovery copy can only be discarded.
- Export writes new files; no format is imported in version 1.
- No data leaves the PC except files the user saves, exports, or prints.

Container, entries, and limits of the document file are in the *File format reference*. The full schema is in architecture section 6.5.

10. External interfaces

10.1 User interface and accessibility

- One window: menu bar, tab strip, page, status bar (see the appendix sketch).
- Keys in the *Keyboard reference*; menus in the *Menus reference* and the *Context menus reference*.
- Accessibility targets are in section 7.4.

10.2 Hardware

Keyboard, mouse, pen, and touch input; any printer with a Windows driver. No other hardware.

10.3 Operating system

Interface	Used for	Through
Windows Open and Save dialogs	Pick files and folders	Electron dialog API
Windows print dialog and spooler	Print	Electron webContents.print
Windows clipboard	Rich text and plain text	Electron clipboard API
Windows spell checker	Underline and suggestions	Electron spellchecker
DPAPI	Encrypt saved signatures	Electron safeStorage
File Explorer	Open .obk files by double-click	MSIX file-type association

10.4 Software libraries

Library	Used for	License	Source
Electron	App runtime	MIT	electronjs.org
React	Interface	MIT	react.dev
ProseMirror	Editor core (proposed)	MIT	prosemirror.net
KaTeX	Equation rendering	MIT	katex.org
mathml2omml	Word equations	LGPL-3.0-or-later	npmjs.com/package/mathml2omml
docx	Word export	MIT	npmjs.com/package/docx
JSZip	Used by docx to package the .docx	MIT (chosen from MIT or GPLv3)	npmjs.com/package/jszip
yauzl	Reading .obk files	MIT	npmjs.com/package/yauzl
yazl	Writing .obk files	MIT	npmjs.com/package/yazl
Mermaid	Diagram block (proposed)	MIT for 12.2.0, checked in spike S-04. Layout is dagre, which is MIT. elkjs is not shipped (APPROVED 27 October 2026). Mermaid still uses DOMPurify 3.4.17 (MPL-2.0 OR Apache-2.0).	mermaid.js.org
Bundled fonts	Text and signatures	See the *Fonts reference*	Each font's license file

UNVERIFIED: mathml2omml, docx, JSZip, yauzl, and yazl were checked on npm. Spike S-04 also checked Mermaid 12.2.0 (MIT) and KaTeX 0.19.1 (MIT). Other licenses are as published by each project and are not yet rechecked. Versions are set in the architecture document.

10.5 Communications

Obelisk makes no network requests itself, and it blocks any request it did not start. Open link hands an http, https, or mailto address to the default browser or mail app after the user confirms; see the *Links reference*. Store updates are handled by Windows, not by Obelisk.

11. Open decisions

Open items use the proposed default until decided. Decided items show the date.

ID	Question	Options	Proposed default	Status
OD-01	Is the app name "Obelisk" free in Partner Center?	Reserve Obelisk; or pick another name	Reserve Obelisk before the manifest is fixed	Open
OD-02	File extension and format version	.obk, format 1; or another extension	.obk, format 1, after checking it is not widely used	Open
OD-03	Autosave interval	15 s (more writes); 30 s; 60 s (more loss)	Within 30 s of the last edit, as a target	Open
OD-04	What does cell math cover?	Arithmetic only; spreadsheet functions; KaTeX in cells	Cell-reference arithmetic only, no named functions	Decided 2026-10-23
OD-05	Diagram engine and syntax	Mermaid (wide use, large); PlantUML (needs Java); none	Mermaid 12.2.0, bundled, offline, dagre layout (MIT). No elkjs. Style post-process and per-render nonce from spike S-04. `click` and `href` are stripped before render.	Decided 2026-10-27. APPROVED 27 October 2026.
OD-06	Should drop order	As listed in section 6	Diagram, drawing pad, signature, Word equations	Decided 2026-10-23
OD-07	Crash safety for never-saved tabs	None; recovery copy in app data	Should: recovery copy every 30 s	Decided 2026-10-23
OD-08	Exact bundled fonts and licenses	Full LibreOffice Latin set; OFL-only subset	LibreOffice Latin set plus Dancing Script, each license checked	Open
OD-09	Theme options	Light and Dark; plus Match Windows	Light, Dark, Match Windows (default)	Open
OD-10	Key choices	Ctrl+Space only; Ctrl+Space plus Ctrl+\	Ctrl+Space, alternate Ctrl+\; Ctrl+Shift+V, Ctrl+Shift+X, Ctrl+U	Decided 2026-10-23
OD-11	Supported Windows versions	Windows 11 only; plus Windows 10 22H2	Windows 11 and Windows 10 22H2, x64	Open
OD-12	Open from File Explorer	Not in 1.0; Should with single instance	Should: one window, new tab per file	Decided 2026-10-23
OD-13	Do table references follow row and column insert and delete?	Rewrite them; keep them static	Rewrite them; a reference to a deleted cell becomes #REF!	Decided 2026-10-23
OD-14	How are other format versions opened?	Refuse every other version; convert older ones	Convert older versions on open, refuse newer ones, and confirm before the first save upgrades a file	Decided 2026-10-23
OD-15	How are saved signatures stored?	A PNG for every kind; strokes or text by kind	A PNG for every kind, one encrypted file each	Decided 2026-10-23
OD-16	A recovery copy neither restored nor	Keep it and offer it again; delete it after some days	Keep it and offer it at every launch	Decided 2026-10-23

ID	Question	Options	Proposed default	Status
	discarded			
OD-17	Restore tabs on launch	Keep, with a session file; drop from 1.0	Keep; open paths are stored in session.json (section 9)	Decided 2026-10-23
OD-18	Animated GIF and WebP images	Keep the animation; keep only the first frame	Keep the original and play it in the editor; PDF, Word, and print use the first frame; Markdown keeps the original file	Decided 2026-10-23
OD-19	Mermaid security level	strict; sandbox; loose	strict, which a diagram cannot change	Decided 2026-10-23
OD-20	Open link	Keep, with a scheme allowlist; remove it	Keep; http, https, and mailto only, after a prompt	Decided 2026-10-23
OD-21	Uncompressed size caps for .obk files	Caps per entry and in total; no caps	document.json 25 MB, each image 20 MB, 250 MB in total	Decided 2026-10-23

12. Acceptance criteria

Each check runs end to end on the reference machine with a packaged build.

ID	Check	Verifies
AC-01	Install as a standard user on both Windows versions; launch to an editable tab within 3 s.	FR-01, NFR-01, NFR-24, NFR-25
AC-02	Build a document with every block type, save, close, and reopen; content, view, and setup match.	FR-07, FR-08, FR-10, FR-11, FR-17, FR-24, FR-32
AC-03	Open each file in the hostile sample set (architecture section 6.10); each opens nothing or opens safely and stays unchanged.	FR-09, NFR-06, NFR-07
AC-04	End the process during 20 large saves; each target holds the old or the complete new file.	NFR-10, NFR-12, NFR-14
AC-05	Edit a saved file for 40 s, end the process, reopen; at most 30 s of edits are lost.	FR-14, NFR-13
AC-06	Change an open file in Notepad, then edit in Obelisk; the conflict choice appears, no autosave runs, and after Cancel the status bar shows Autosave paused.	FR-04, FR-13, FR-15
AC-07	Close a changed tab and exit with one; each asks to save, discard, or cancel.	FR-03, FR-16
AC-08	A new file opens in Pages view; switching marks it changed; the PDF from Pageless has pages.	FR-22, FR-23, FR-62
AC-09	Create an MLA paper and compare its PDF with the *MLA template reference*.	FR-25, FR-27, FR-28, FR-29, FR-30
AC-10	Type each sample equation from the *Equation reference*; each renders live, and a broken one shows an error.	FR-53, FR-54, FR-55

ID	Check	Verifies
AC-11	Export the equation sample to Word with conversion on, then off; equations edit, then show as images.	FR-56, FR-57, FR-63
AC-12	Export a document with images to Markdown and view it on GitHub; images and tables show.	FR-52, FR-64, FR-65
AC-13	Print and export a changed document; the .obk bytes are unchanged and replace prompts appear.	FR-12, FR-61, FR-66
AC-14	Run every example in the *Table math reference*, then insert and delete referenced rows; values, references, errors, and sort order match.	FR-45, FR-46, FR-47, FR-48
AC-15	Select, move, duplicate, and copy blocks between tabs and into Word and Notepad.	FR-33, FR-38, FR-39, FR-40, FR-41, FR-42
AC-16	Run the new-document and export use cases by keyboard only, with Narrator on.	FR-02, FR-34, FR-35, FR-43, FR-44, NFR-15, NFR-16
AC-17	Check contrast, 200% scaling, and color-only signals in Light and Dark.	NFR-17, NFR-18, NFR-19
AC-18	Replace a word used five times with Whole word on; one undo restores all five.	FR-58, FR-59, FR-60
AC-19	Insert each image type, resize, rotate, flip, and try one over each limit, including an animated GIF with too many frames.	FR-49, FR-50, FR-51
AC-20	Run a full session with spelling on, an equation, a diagram, and every export, under a packet capture; no traffic leaves the PC. The session does not use Open link.	NFR-05
AC-21	Should: double-click two .obk files, with Obelisk closed and then running; both open as tabs in one window.	FR-18
AC-22	Should: end the process with a changed never-saved tab, restart, restore, and save; the copy is gone.	FR-19, FR-20, FR-21
AC-23	Should: insert a date and time item, try each format, and reopen the file a day later; the value is unchanged.	FR-67, FR-68, FR-69
AC-24	Should: build rows of 2, 3, and 4 columns, resize, unsplit one, and print a row across a page end.	FR-70, FR-71, FR-72, FR-73, FR-74
AC-25	Should: draw on a pad and over an image, rotate the image, and export to PDF and Word.	FR-75, FR-76, FR-77
AC-26	Should: create each kind of signature, reuse a saved one, set its options, and export to each format.	FR-78, FR-79, FR-80, FR-81, NFR-08, NFR-09
AC-27	Should: render a flowchart offline, then break its source; the error shows in place.	FR-82, FR-83
AC-28	Pass Store certification; the About page lists every notice; uninstall keeps documents.	NFR-26, NFR-27, NFR-28, NFR-29, NFR-30, NFR-31
AC-29	Open link on an https link shows the full address and asks first; file:, javascript:, and network-path links are refused on open and paste.	FR-44, NFR-05, NFR-06
AC-30	Insert an animated GIF; it plays in the editor and stays still with Windows animations off. PDF, Word, and print show the first frame; Markdown keeps the GIF.	FR-49, FR-62, FR-63, FR-65, NFR-20
AC-31	Turn on Restore tabs, open two files, exit, delete one file, and relaunch; the other reopens and one message names the missing file.	FR-05, FR-06

13. Future features

These are not requirements. Each is deferred for the reason given.

Feature	Reason deferred
Export to HTML and plain text (.txt)	Kept out to finish PDF, Word, and Markdown well first.
Imports (Word, Markdown, Polaris .plr)	Import fidelity is a large job; 1.0 starts clean at format 1.
More templates (APA, Chicago, letter)	MLA covers the main student need.
A4 and other paper sizes	Letter covers the first users.
Different first page for headers	Dropped from 1.0 to keep page setup simple.
macOS and Linux builds	Release is Microsoft Store only.
Verified digital signatures	Needs certificates and a trust model; the signature block is visual only.

14. Traceability

Each requirement maps to the use cases and acceptance checks that cover it. Architecture elements are mapped in the architecture document.

Requirement	Priority	Use cases	Acceptance	Method
FR-01	Must	UC-02	AC-01	Demo
FR-02	Must	UC-02	AC-16	Inspection
FR-03	Must	UC-02	AC-07	Test
FR-04	Must	UC-02, UC-03	AC-06	Test
FR-05	Must	UC-01	AC-31	Inspection
FR-06	Must	UC-01	AC-31	Test
FR-07	Must	UC-02	AC-02	Test
FR-08	Must	UC-03	AC-02	Test
FR-09	Must	UC-01, UC-03, UC-10	AC-03	Test
FR-10	Must	UC-02	AC-02	Test
FR-11	Must	UC-02	AC-02	Demo
FR-12	Must	UC-02, UC-05	AC-13	Test
FR-13	Must	UC-03	AC-06	Test
FR-14	Must	UC-03	AC-05	Test
FR-15	Must	UC-03	AC-06	Test
FR-16	Must	UC-03	AC-07	Test
FR-17	Must	UC-01, UC-03	AC-02	Test
FR-18	Should	UC-01	AC-21	Demo
FR-19	Should	UC-10	AC-22	Inspection
FR-20	Should	UC-10	AC-22	Test
FR-21	Should	UC-10	AC-22	Inspection
FR-22	Must	UC-02	AC-08	Test
FR-23	Must	UC-03	AC-08	Test
FR-24	Must	UC-04	AC-02	Test
FR-25	Must	UC-04	AC-09	Test
FR-26	Must	UC-04	-	Test
FR-27	Must	UC-04	AC-09	Inspection
FR-28	Must	UC-03	AC-09	Test
FR-29	Must	UC-03	AC-09	Test
FR-30	Must	UC-04	AC-09	Test
FR-31	Must	UC-04	-	Demo
FR-32	Must	UC-02	AC-02	Test
FR-33	Must	UC-02	AC-15	Demo
FR-34	Must	UC-02, UC-07	AC-16	Test
FR-35	Must	UC-02	AC-16	Test
FR-36	Must	UC-06	-	Test
FR-37	Must	UC-02, UC-06	-	Test
FR-38	Must	UC-06	AC-15	Test

Requirement	Priority	Use cases	Acceptance	Method
FR-39	Must	UC-06	AC-15	Test
FR-40	Must	UC-06	AC-15	Test
FR-41	Must	UC-06	AC-15	Test
FR-42	Must	UC-06	AC-15	Test
FR-43	Must	UC-06	AC-16	Test
FR-44	Must	UC-06	AC-16, AC-29	Inspection
FR-45	Must	UC-07	AC-14	Test
FR-46	Must	UC-07	AC-14	Test
FR-47	Must	UC-07	AC-14	Test
FR-48	Must	UC-07	AC-14	Test
FR-49	Must	UC-04	AC-19, AC-30	Test
FR-50	Must	UC-04	AC-19	Test
FR-51	Must	-	AC-19	Test
FR-52	Must	UC-04	AC-12	Inspection
FR-53	Must	UC-04	AC-10	Test
FR-54	Must	UC-04	AC-10	Demo
FR-55	Must	UC-04	AC-10	Test
FR-56	Must	UC-05	AC-11	Test
FR-57	Should	UC-05	AC-11	Test
FR-58	Must	UC-08	AC-18	Test
FR-59	Must	UC-08	AC-18	Test
FR-60	Must	UC-08	AC-18	Test
FR-61	Must	UC-05	AC-13	Demo
FR-62	Must	UC-04, UC-05	AC-08, AC-30	Test
FR-63	Must	UC-05	AC-11, AC-30	Test
FR-64	Must	UC-05	AC-12	Test
FR-65	Must	UC-05	AC-12, AC-30	Test
FR-66	Must	UC-05	AC-13	Test
FR-67	Should	UC-09	AC-23	Test
FR-68	Should	-	AC-23	Test
FR-69	Should	-	AC-23	Test
FR-70	Should	UC-06	AC-24	Test
FR-71	Should	UC-06	AC-24	Test
FR-72	Should	UC-06	AC-24	Test
FR-73	Should	-	AC-24	Test
FR-74	Should	-	AC-24	Test
FR-75	Should	UC-09	AC-25	Test
FR-76	Should	UC-04	AC-25	Test
FR-77	Should	-	AC-25	Inspection
FR-78	Should	UC-09	AC-26	Test
FR-79	Should	UC-09	AC-26	Test

Requirement	Priority	Use cases	Acceptance	Method
FR-80	Should	UC-09	AC-26	Inspection
FR-81	Should	UC-09	AC-26	Test
FR-82	Should	-	AC-27	Test
FR-83	Should	-	AC-27	Test
NFR-01	Must	-	AC-01	Test
NFR-02	Must	-	-	Test
NFR-03	Must	-	-	Test
NFR-04	Must	UC-05	-	Test
NFR-05	Must	-	AC-20, AC-29	Analysis
NFR-06	Must	-	AC-03, AC-29	Test
NFR-07	Must	-	AC-03	Test
NFR-08	Should	UC-09	AC-26	Inspection
NFR-09	Should	UC-09	AC-26	Test
NFR-10	Must	UC-03	AC-04	Test
NFR-11	Must	UC-02	-	Test
NFR-12	Must	UC-02	AC-04	Test
NFR-13	Must	UC-03	AC-05	Test
NFR-14	Must	-	AC-04	Test
NFR-15	Must	-	AC-16	Test
NFR-16	Must	-	AC-16	Test
NFR-17	Must	-	AC-17	Analysis
NFR-18	Must	-	AC-17	Inspection
NFR-19	Must	-	AC-17	Inspection
NFR-20	Should	-	AC-30	Demo
NFR-21	Must	-	-	Inspection
NFR-22	Must	-	-	Inspection
NFR-23	Should	-	-	Analysis
NFR-24	Must	UC-01	AC-01	Test
NFR-25	Must	UC-01	AC-01	Test
NFR-26	Must	-	AC-28	Inspection
NFR-27	Must	-	AC-28	Test
NFR-28	Must	-	AC-28	Inspection
NFR-29	Must	-	AC-28	Inspection
NFR-30	Must	-	AC-28	Inspection
NFR-31	Must	-	AC-28	Demo

References

Every list in the requirements points to one of these named tables.

Menus reference

Each item is enabled only when it applies; for example, Paste is disabled with an empty clipboard.

An item that needs more input before it acts ends with an ellipsis (…).

Menu	Items
File	New; New from template > MLA; Open…; Save; Save as…; Print…; Export > PDF…, Word…, Markdown…; Close tab; Exit
Edit	Undo; Redo; Cut; Copy; Paste; Paste as plain text; Duplicate; Select all; Find; Replace
View	Pages; Pageless; Zoom in; Zoom out; Zoom to 100% (50% to 200%, 10% steps)
Insert	Every block in the *Slash menu reference*, in its order and groups; Date and time… replaces Date; Link… (Ctrl+K) follows it in group 5.
Format	Text; Font; Font size; Paragraph…; Increase indent; Decrease indent; Clear formatting; Page setup…; Header and footer
Tools	Spelling on or off; Settings
Help	Keyboard shortcuts; About Obelisk

Keyboard reference

Menus also have Alt access keys. Top menus: File F, Edit E, View V, Insert I, Format O, Tools T, Help H. Inside a table, Tab and Shift+Tab move between cells.

Where	Keys	Action
Anywhere	Ctrl+N, Ctrl+O	New document, Open
Anywhere	Ctrl+S, Ctrl+Shift+S	Save, Save as…
Anywhere	Ctrl+P, Ctrl+Shift+E	Print, Export
Anywhere	Ctrl+W	Close tab
Anywhere	Ctrl+Tab, Ctrl+Shift+Tab	Next tab, previous tab
Anywhere	Ctrl+comma	Settings
Anywhere	Ctrl+F, Ctrl+H	Find, Replace
Anywhere	F6	Move focus: menu bar, tabs, page, status bar
Anywhere	Ctrl+plus, Ctrl+minus, Ctrl+0	Zoom in, zoom out, 100%
Editor	Ctrl+Z, Ctrl+Y	Undo, Redo (100 steps)
Editor	Ctrl+B, Ctrl+I, Ctrl+U	Bold, italic, underline
Editor	Ctrl+Shift+X	Strikethrough
Editor	Ctrl+K	Link
Editor	Ctrl+Space, or Ctrl+\	Clear formatting
Editor	Ctrl+L, Ctrl+E, Ctrl+R, Ctrl+J	Align left, center, right, justify
Editor	Ctrl+Shift+>, Ctrl+Shift+<	Larger, smaller font size

Where	Keys	Action
Editor	Tab, Shift+Tab	Increase indent, Decrease indent
Editor	Ctrl+Enter	Page break
Editor	Ctrl+A	Select all
Editor	Ctrl+X, Ctrl+C, Ctrl+V	Cut, copy, paste (text or blocks)
Editor	Ctrl+Shift+V	Paste as plain text
Editor	Esc	Select the current block
Editor	Shift+F10, Menu key	Open the context menu
Selected blocks	Ctrl+D	Duplicate
Selected blocks	Alt+Shift+Up, Alt+Shift+Down	Move up, move down

Typed shortcuts

Typed shortcuts follow GitHub Flavored Markdown and are always on.

Type	Where	Result
`# to ###### and a space`	Line start	Heading 1 to 6
`- or * and a space`	Line start	Bulleted list
`1. and a space`	Line start	Numbered list
`[ ] and a space`	Line start	Checklist
`> and a space`	Line start	Quote
``	Line start	Code block
`---`	Line start	Horizontal line
`$$`	Line start	Equation block
`**text**`	Inline	Bold
`*text*`	Inline	Italic (asterisks only; underscores stay as typed)
`~~text~~`	Inline	Strikethrough
`` `text` ``	Inline	Inline code
`$...$`	Inline	Inline equation

- An inline mark applies when its closing mark is typed.
- Ctrl+Z right after a shortcut fires restores the typed characters.
- Block shortcuts fire only at the start of a line.
- No shortcut fires inside a code block or an equation block.

UNVERIFIED: Ctrl+Space can switch the input method on some Windows keyboard layouts (for example Chinese IMEs); Ctrl+\ is the alternate. Ctrl+Shift+> and < depend on the layout. Check both as Electron menu accelerators on a non-US layout.

Context menus reference

Composition rule: every block shows the block-handle menu plus its own items. Every inline item shows the text menu plus its own items.

Menu	Base	Own items
Text	-	Spelling suggestions; Cut; Copy; Paste; Paste as plain text; Link…; Clear formatting
Block handle	-	Cut; Copy; Duplicate; Delete; Turn into; Insert above; Insert below; Move up; Move down
Table cell	Block handle	Insert row above or below; Insert column left or right; Delete row; Delete column; Sort ascending; Sort descending; Header row
Image	Block handle	Alt text…; Rotate left; Rotate right; Flip horizontal; Flip vertical; Draw on image; Reset size; Replace…
Equation block	Block handle	Edit LaTeX; Copy LaTeX
Column row	Block handle	Add column; Remove column; Even widths
Drawing pad	Block handle	Clear drawing; Reset size
Diagram	Block handle	Edit source; Copy source
Signature	Block handle, minus Turn into	Change signature…; Edit line and labels…; Blank mode; Reset size
Link	Text	Open link (asks first; see the *Links reference*); Edit link…; Remove link
Inline equation	Text	Edit LaTeX; Copy LaTeX
Date and time	Text	Edit date and time…; Change format; Convert to text; Remove

Slash menu reference

Typing after / filters the list by name or search word. Enter inserts; Esc closes and keeps the typed text.

Items are listed in this order, in six groups with a separator between groups.

Item	Search words	Priority
Text	paragraph, p	Must
Heading 1, 2, 3	h1, h2, h3, title	Must
Bulleted list, Numbered list, Checklist	ul, ol, todo	Must
Quote, Code block	blockquote, code	Must
Columns	column, side by side	Should
Table	grid, tab	Must
Image	picture, photo, img	Must
Equation	equation, math, latex, formula	Must
Diagram	mermaid, flowchart, chart	Should
Drawing	draw, sketch, pad	Should
Signature	sign, signature	Should
Date	date, time, today, now	Should
Horizontal line	hr, line, divider	Must
Page break	break, new page	Must

Block types reference

Block	Holds	Priority
Paragraph	Formatted text, links, inline equations, date and time items	Must
Heading 1 to 6	Formatted text	Must
Bulleted, numbered, checklist	List items, nested up to 6 levels; checklist items have a check state	Must
Quote	Formatted text	Must
Code block	Plain text in Consolas; no shortcuts	Must
Horizontal line	Nothing	Must
Table	Cells of formatted text or formulas; optional header row; up to 200 rows and 26 columns	Must
Image	One image, alt text, size, rotation, flip; Should: drawing layer	Must
Equation block	LaTeX source; shown rendered as display math	Must
Page break	Nothing; forces a new page in Pages view and print	Must
Column row	Two to four blocks side by side; no column rows or page breaks inside	Should
Drawing pad	Strokes	Should
Signature	Signature image, line, labels, blank mode	Should
Diagram	Mermaid source; shown rendered	Should

Formatting reference

Kind	Options
Text (character marks)	Bold, italic, underline, strikethrough, superscript, subscript, inline code, text color, highlight
Font	Any font in the *Fonts reference*; default Liberation Serif
Font size	8, 9, 10, 11, 12, 14, 16, 18, 20, 24, 28, 36, 48, 72 pt; default 12 pt
Alignment	Left, center, right, justify
Line spacing	Single, 1.15, 1.5, double
Space before and after	0, 6, 12, 18, 24 pt
Indents	Left 0 to 3 in; first line 0 to 1 in; hanging 0 to 1 in; steps of 0.25 in. Increase indent and Decrease indent move the left indent one 0.25 in step.
Clear formatting	Removes Text formatting, font, and font size; keeps the block type and links

Fonts reference

Bundled fonts are the LibreOffice Latin set plus Dancing Script (for typed signatures). The licenses below are expected, not yet checked.

Font family	Use	Expected license
Liberation Serif, Sans, Mono	Default text; metric-compatible with Times New Roman, Arial, Courier New	SIL OFL 1.1
Caladea, Carlito	Metric-compatible with Cambria and Calibri	SIL OFL 1.1

Font family	Use	Expected license
DejaVu Serif, Sans, Sans Mono	Wide character coverage	Bitstream Vera-based DejaVu license (not OFL)
Linux Libertine G, Linux Biolinum G	Book text and headings	SIL OFL or GPL with font exception
Noto Serif, Sans (Latin)	Fallback text	SIL OFL 1.1
Dancing Script	Typed signatures	SIL OFL 1.1

A file that names a font Obelisk does not bundle still opens. The text shows in the default font, the status bar says so once, and the file keeps the font name.

UNVERIFIED: The exact LibreOffice Latin set and each license (OD-08). Check each font's license file from its upstream source before release.

Page setup reference

Setting	Values
Paper	Letter (8.5 by 11 in) only
Orientation	Portrait (default), landscape
Margins	Normal 1 in (default); Narrow 0.5 in; Wide 1.5 in left and right; Custom 0.25 to 3 in per side; text width at least 2 in
Header and footer	One paragraph of formatted text with page fields, on every page, 0.5 in from the edge; hidden in Pageless view
Header and margins	The header and footer reserve their own space, so the body starts below the header whatever the margin
Fields	Page number; page count
Page number formats	1, 2, 3; i, ii, iii; Page 1 of N; set on each field
Not offered	Different first page; odd and even pages; sections

MLA template reference

Setting	Value
Margins	1 in on all sides
Font	Liberation Serif, 12 pt
Line spacing	Double, everywhere, with no extra space before or after
Header	Right-aligned: last name, a space, and the page number
Heading lines	Four left-aligned placeholder lines: name, instructor, course, date
Title	Centered placeholder line, no bold or underline
Body	First-line indent 0.5 in
Works Cited	New page, centered title, hanging indent 0.5 in

UNVERIFIED: Check each row against the MLA Handbook, 9th edition, before release.

Paste reference

From other apps	Kept	Dropped
Text	Paragraphs, headings, lists, bold, italic, underline, strikethrough, links (http, https, mailto)	Fonts, colors, sizes, styles, classes; other link schemes
Tables	Cells and text	Widths, borders, cell colors, merged cells (split)
Images	Images on the clipboard as image data, within the *Image reference* limits	Images referenced by URL; SVG
Code and markup	Shown as text	Scripts, iframes, forms, event handlers
Paste as plain text	Text and line breaks	Everything else

Pasted HTML is read without loading anything, then mapped into Obelisk blocks; raw HTML is never inserted.

Pasting Obelisk blocks between tabs keeps their content exactly. They are checked like an opened file and get new block IDs.

UNVERIFIED: Two claims are unchecked: that HTML parsed into a detached document loads no images and runs no scripts in Electron, and how Electron reads a custom clipboard format on Windows. Check both with a test that watches the network.

Table math reference

A cell whose text starts with = holds a formula. Formulas are cell-reference arithmetic only; there are no named functions.

Rule	Detail
References	A1 style: column letter and row number; the header row, if any, is row 1
Numbers	Decimal numbers such as 3, -2.5, 0.75; no thousands separators
Operators	+ − * /, and power as ** or ^
Precedence	Power first (right to left), then * and /, then + and -; -2^2 is -4
Grouping	( ) or [ ], nested to any depth; each must close with its own kind
Empty cell	Counts as 0
#REF!	A reference outside the table or to a text cell; a reference to a deleted row or column becomes #REF! in the formula
Insert and delete	Inserting or deleting a row or column rewrites references to cells that move, in the same undo step
Size	Up to 200 rows and 26 columns (A to Z)
#DIV/0!	Division by zero
#CYCLE!	A formula that depends on itself, directly or through other cells
Does not parse	Kept as text with an error underline; for example =B2+ stays text
Display	The formula while the cell is edited; otherwise the value, up to 10 decimals
Recalculation	After every edit, in dependency order
Sort	One column, ascending or descending; header row fixed; numbers before text; empty cells last; one undo step

Rule	Detail
Sort and formulas	Formulas keep their text; references are not rewritten

Example	Result
`=B2+B3*2`	B2 plus twice B3
`=(B2-B3)/2`	Half the difference
`=[B2+B3]**2`	Square of the sum
`=2^3^2`	512 (right to left)
`=B2/0`	#DIV/0!
`=Z99`	#REF!
`=B4`	Becomes =B5 when a row is inserted above row 4
`=B3`	Becomes =#REF! when row 3 is deleted

Image reference

Rule	Value
Types	PNG, JPEG, GIF, WebP, recognized by their first bytes; SVG is refused
Animated GIF and WebP	Kept as is and played in the editor; shown still on the first frame when Windows animation effects are off (OD-18)
Animated in exports	PDF, Word, and print use a PNG of the first frame, made in the sandboxed renderer at export time (spike S-06). Markdown keeps the original file
Largest file	20 MB per image
Largest size	8000 by 8000 pixels, read from the image header before decoding
Animation limits	Up to 1,000 frames and 250 million pixels across all frames (targets); each frame inside the image size
Checked	On insert, paste, and open; a file with an image over a limit is refused
Stored	Original bytes inside the document file, named by content hash; stored once when several blocks use it
Default width	The smaller of the image width and the text width
Rotate and flip	Quarter turns; horizontal and vertical flip; stored, not baked in
Alt text	Up to 500 characters

Equation reference

Rule	Detail
Syntax	The LaTeX math that KaTeX supports (katex.org/docs/supported.html)
Block	$$ at a line start, or /equation; display math, centered
Inline	Between single $ signs inside text
Errors	KaTeX's message, in red, in place of the equation
Not supported	\href, \url, \includegraphics, and HTML extensions; macros carried from one equation to another; sizes over 50 em
Limits	Up to 10,000 characters of LaTeX per equation

Rule	Detail
PDF and print	Rendered KaTeX
Word	OMML equation (Should); image fallback (Must)
Markdown	$…$ and $$…$$ with the LaTeX source

Sample	LaTeX
Fraction	`\frac{a+b}{c}`
Root and power	`\sqrt{x^2+y^2}`
Sum	`\sum_{i=1}^{n} i^2`
Integral	`\int_0^1 x\,dx`
Matrix	`\begin{pmatrix}1&0\\0&1\end{pmatrix}`
Greek and accents	`\hat{\theta} = \alpha + \beta`

Export reference

Block	PDF	Word (.docx)	Markdown (.md)
Text and headings	As on the page	Paragraph and heading styles	Text, # headings
Lists and checklist	As on the page	Word lists. Checkboxes are an option: a read-only Segoe UI Symbol mark, or a clickable Word checkbox content control.	-, 1., - [ ]
Quote, code, Horizontal line	As on the page	Quote style; Consolas; rule	>, ```, ---
Table	As on the page; values	Word table; values	GFM table; values
Image	Image; drawing flattened; first frame if animated	Image with alt text; drawing flattened; first frame if animated	Original file in the images folder; alt text
Equation	Rendered	OMML (Should) or image	$ and $$ LaTeX
Page break	New page	Page break	Ignored
Header and footer	On every page	Word header and footer	Ignored
Date and time	Formatted text	Formatted text	Formatted text
Column row	Side by side	Table without borders	Stacked in order
Drawing, signature, diagram	Image	Image	PNG file in the images folder

- Markdown export escapes text, alt text, and link text, so nothing in the document becomes Markdown or HTML markup. Word export writes text only as text.
- Markdown images go to a folder named after the .md file plus _images, as image-001.png, image-002.gif, and so on. Names never come from the document.
- An existing images folder is part of the replace prompt (FR-12). Obelisk replaces only the files it writes and never deletes others.
- If an export fails or stops, no partial file is left; a file being replaced keeps its old bytes.

- Spike S-05 checked an existing `.obk` name with a Word `.docx` filter. Windows kept the `.obk` name and asked to replace that file. The filter did not change the extension. Export sets the extension itself. It does not rely on the filter to turn a name into `.docx` or `.obk`.
- Before Word text, alt text, or link text is written, XML 1.0-illegal control characters are removed. docx escapes <, &, and `]]>`. Unpaired surrogates do not survive as surrogates. Spike S-09: Word and LibreOffice both refused a file that still contained those control characters.
- Exported pictures of equations, drawings, and diagrams are cropped to their content, with a small margin, and rendered at 2× or more.
- The Word exporter sets Letter page size and the margins itself. The docx library's default page is A4. It also sets the document's font and each table's width. It does not leave those to the library. The spike files used Calibri 11 pt only because they were not Obelisk documents. Obelisk uses the document font, Liberation Serif 12 pt unless the user changed it.

Date formats reference

Examples show 9:05 PM on March 4, 2027.

Format	Example
Windows short date (default)	3/4/2027
Windows long date	Thursday, March 4, 2027
Month day, year	March 4, 2027
Day month year (MLA)	4 March 2027
ISO date	2027-03-04
Short date and time	3/4/2027 9:05 PM
Time only	9:05 PM

UNVERIFIED: Reading the Windows date format from Electron (app.getLocale and Intl) may differ from the user's Region settings. Check on a PC with custom Region formats.

Columns reference

Rule	Value
Columns per row	2 to 4; no columns inside a column
New split	Even widths
Smallest share	15% of the text width
Gap	0.25 in
Even widths	Context menu item; resets shares
Paging	A row that fits on a page but not in the space left moves whole to the next page
Tall rows	A row taller than a page splits at the page bottom, each column continuing on the next page
Pageless view	Side by side, no paging

Drawing tools reference

Tool	Options
Pen	Thin, medium, thick; 8 colors
Highlighter	Thick, translucent; 4 colors
Eraser	Erases whole strokes
Shapes	Line, arrow, rectangle, ellipse
Undo	Ctrl+Z within the pad, as editor undo steps
Input	Mouse, pen, touch

Signature options reference

Option	Values
Create by	Draw; type in Dancing Script; image file
Line	None, or a line below the signature
Width	1.5 to 4 in
Labels	Up to two lines below the line, for example a name and a title
Alignment	Left, center, right
Blank mode	Shows only the line and labels, for signing on paper
Saved signatures	Up to 10, each a PNG with a name of 1 to 40 characters, unique ignoring case; delete from the block menu
Deleting a saved signature	Leaves documents that already use it unchanged

Settings reference

Setting	Values	Default
Theme	Light, Dark, Match Windows	Match Windows
Spelling	On, off	On
Default font	Any font in the *Fonts reference*	Liberation Serif
Default font size	Any size in the *Formatting reference*	12 pt
Restore tabs on launch	On, off	Off
Show page edges in Pages view	On, off	On

Not settings, by design: the view for new documents, the undo depth, and Markdown shortcuts.

Settings are checked when read; a bad or unknown value falls back to its default. With Restore tabs on launch on, the open files are kept in session.json in app data (section 9). Each reopens through the usual file checks, and missing files are named in one message.

Links reference

Rule	Value
Schemes	http, https, and mailto; any other is refused, including file:, javascript:, and network paths

Rule	Value
Checked	On open, paste, and Ctrl+K, and again just before the link is opened
Open link	Shows the full address and asks first; then the default browser or mail app opens it
Length	Up to 2,048 characters

File format reference

Rule	Value
Extension	.obk (OD-02)
Container	ZIP archive; stored or deflated entries only; no encryption, no archives inside
Entries	Exactly `manifest.json`, `document.json`, and one `images/<sha256>.<ext>` per image (png, jpg, gif, webp); any other name is refused
Manifest	Format name obelisk, format version, writer version, and the image list (hash, type, bytes, pixel size), which must match the images exactly
Format version	1, a whole number. Older versions are converted on open, and the first save asks before upgrading the file. Newer versions are refused (OD-14)
Schema	document.json matches the schema in architecture section 6.5; unknown block types or fields are refused
Largest file	200 MB
Largest entry count	2,000 entries
Largest uncompressed size	document.json 25 MB; each image 20 MB; 250 MB in total (OD-21)
Other limits	Blocks, text, tables, drawings, and sources, as listed in architecture section 6.6
Paths	Only the entry names above; read in memory, never unpacked to disk; every image is used by a block
Text	UTF-8 JSON; duplicate keys refused; no HTML is stored or rendered
Checked	On open, on restoring a recovery copy or session, and on pasting Obelisk blocks

Checked in spike S-08 (yauzl 3.4.0, yazl 3.3.1, Electron 45.0.0, Windows image 20261021.278.1). Hostile samples H-01 to H-09 were refused. That includes a bomb whose headers claim a tiny size, stopped at the real-byte caps. Bad names were refused: `../`, a leading slash, a backslash, drive letters, network paths, stream suffixes, `CON`, and case duplicates. Two saves of a 200-image document were the same bytes. A refused file was unchanged (FR-09).

Decoding runs in a utility process. Electron 45 does not OS-sandbox that process. The checks above are the defence. ADR-013 stays. APPROVED 27 October 2026: the main-process fallback is not taken. No FR is cut.

Open-source license reference

Component	License	Notice shown
Electron and Chromium	MIT; Chromium notices	About, full text
React	MIT	About
ProseMirror	MIT	About

Component	License	Notice shown
KaTeX	MIT	About
mathml2omml	LGPL-3.0-or-later	About, full text, source link
docx	MIT	About
JSZip (used by docx)	MIT (chosen from its MIT or GPLv3 dual license)	About
yauzl	MIT	About
yazl	MIT	About
Mermaid	MIT for 12.2.0 (spike S-04). Layout is dagre (MIT). elkjs is not shipped.	About
Bundled fonts	Per the *Fonts reference*	About, each license file

UNVERIFIED: What LGPL-3.0 requires for mathml2omml packed in an Electron app (for example inside an asar archive in an MSIX). Check the LGPL-3.0 text and github.com/fiduswriter/mathml2omml.

Appendix: Interface sketch

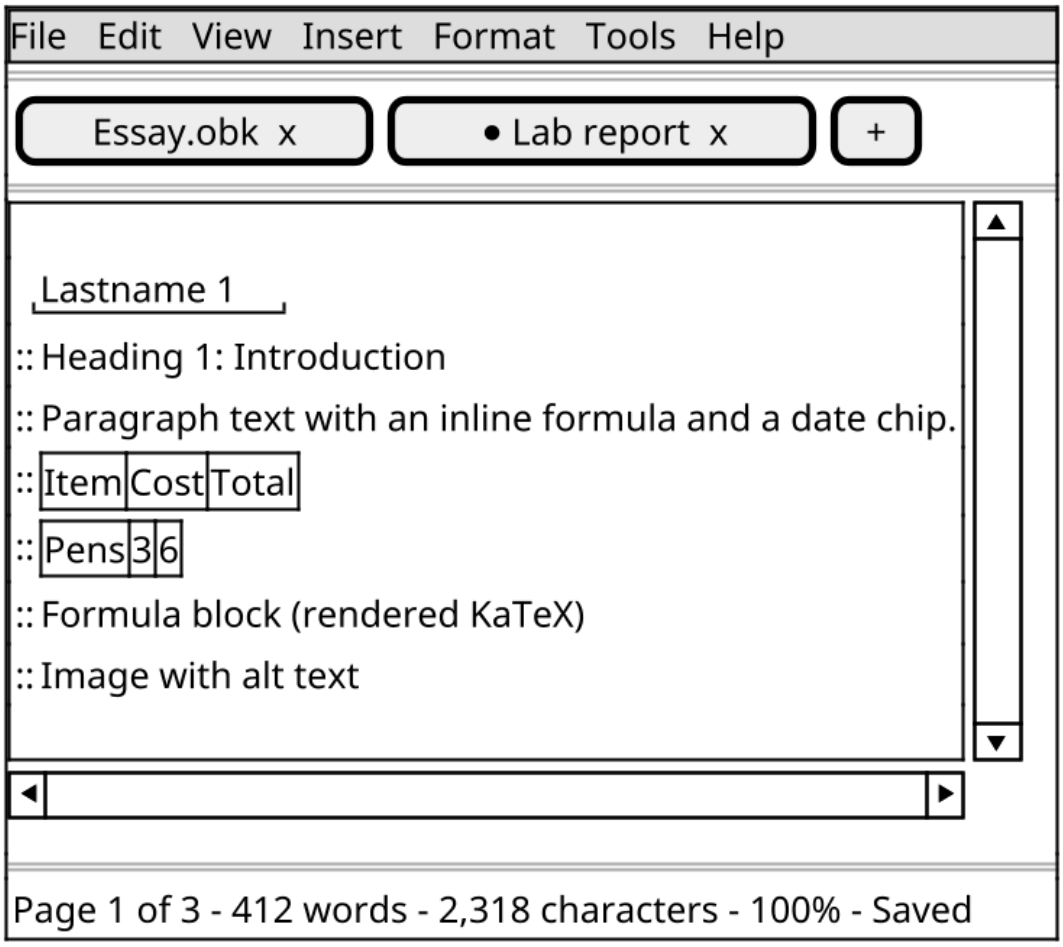


Figure 2. *Editor window sketch*

Purpose: Shows the window regions named in FR-01.

Assumption: Pages view at 100% zoom with two tabs open.

Limitation: A wireframe; spacing and styling are not final.